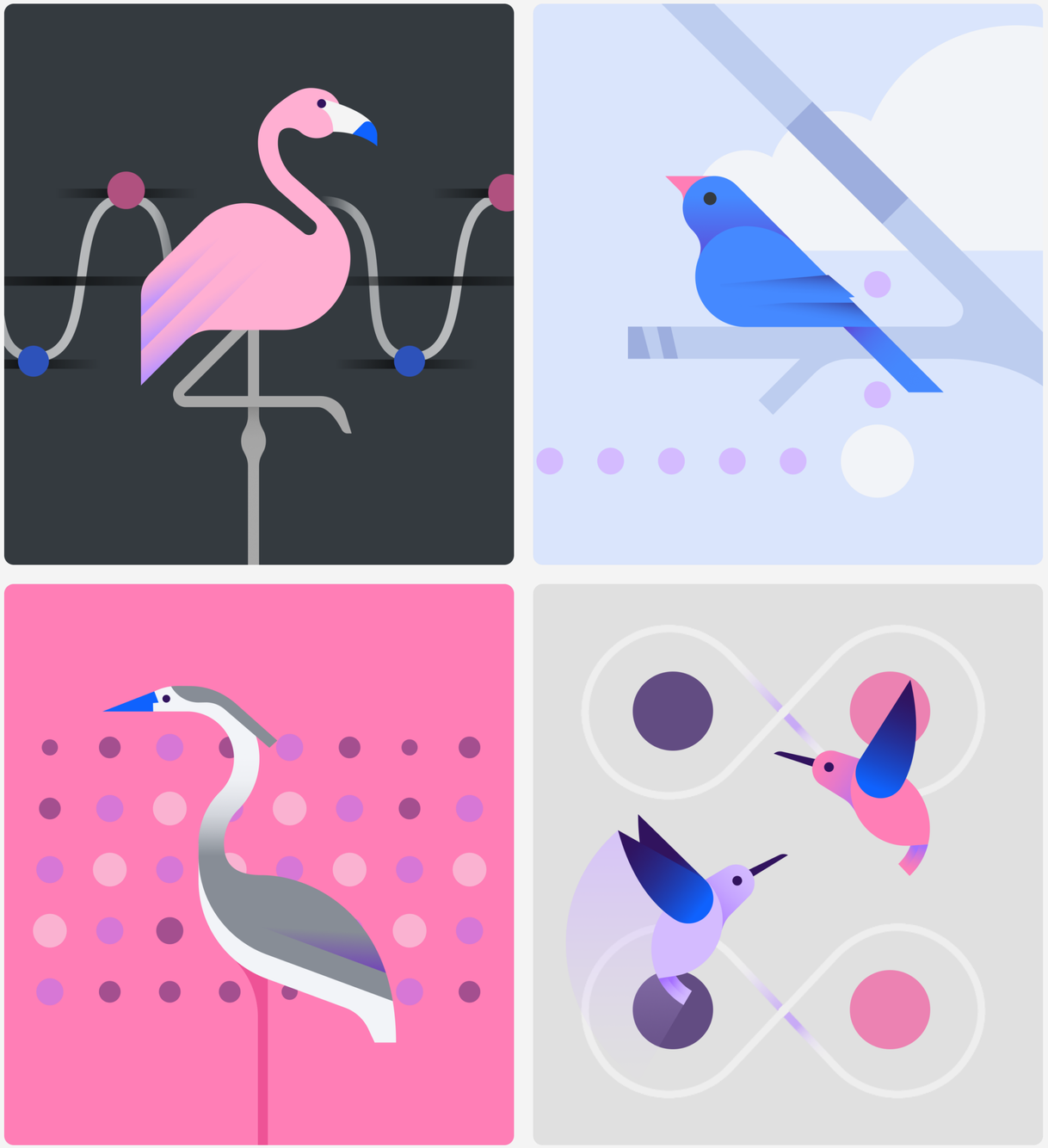
Qiskit Fall Fest 2026

Quantum Bootcamp

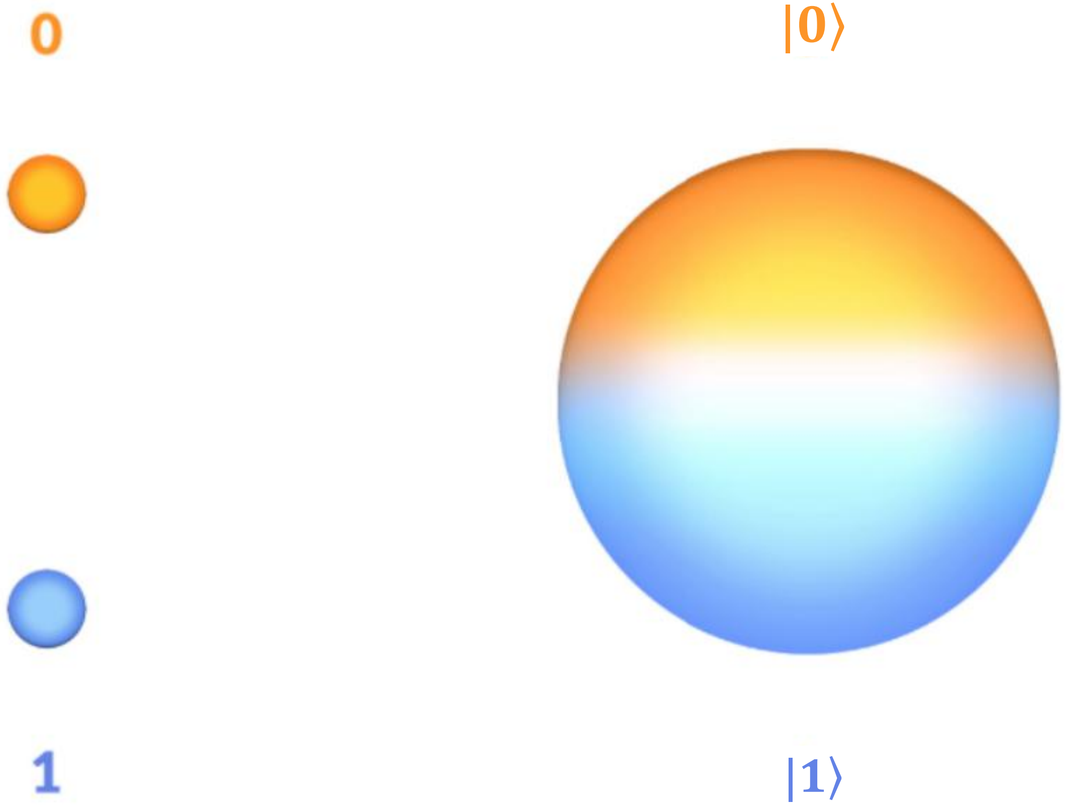


Bootcamp* Plan

1	2	3	4
From classical to quantum information. Notation.	Single-qubit operations. Superposition.	Multi-qubit operations. Entanglement.	Quantum circuits.

*Based on **Basics of quantum information course** on IBM Quantum Platform:
https://quantum.cloud.ibm.com/learning/en/courses/basics-of-quantum-information?lp=introduction-to-quantum

From classical to quantum information. Notation.

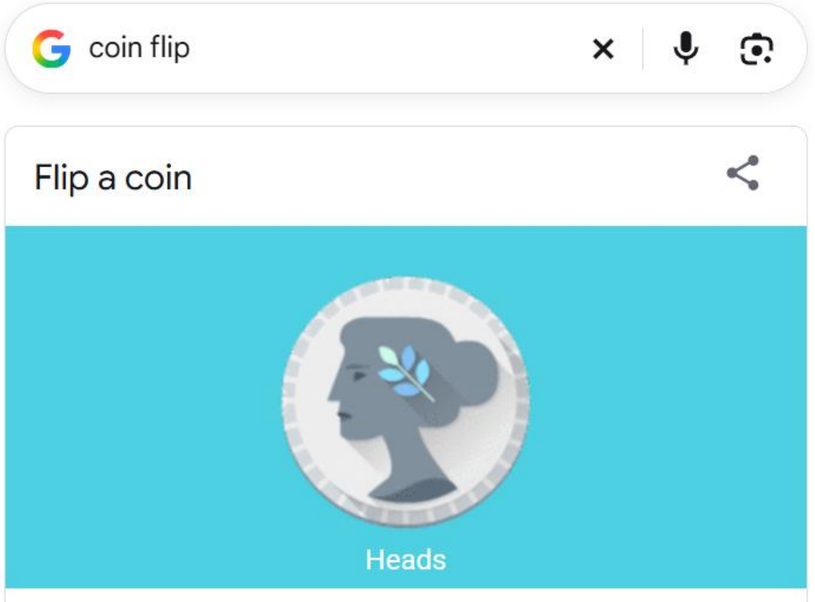


Information representation

Consider a physical system that stores information: let us call it X.



Assume X can be in one of a finite number of **classical states** at each moment.
Denote this **classical state set by Σ**.



Examples:

- If X is a bit, then its classical state set is $\Sigma = \{0, 1\}$.
- If X is a fair coin, then its classical state set is $\Sigma = \{\text{heads}, \text{tails}\}$.
- If X is a six-sided die, then $\Sigma = \{1, 2, 3, 4, 5, 6\}$.

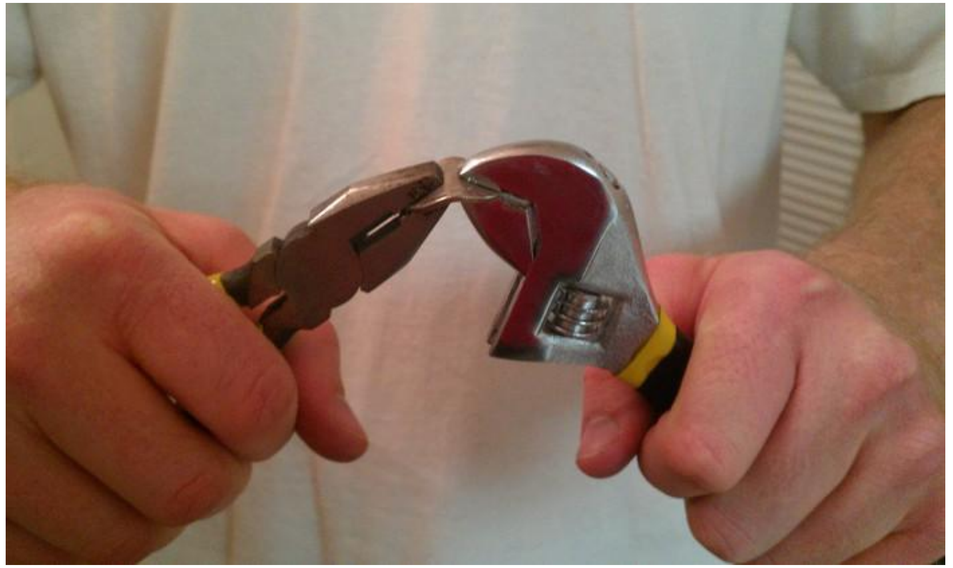
There may be **uncertainty** about the classical state of a system, where each classical state has some **probability** associated with it.

Probability of a classical state

For example,

if X is **an unfair coin**, then, perhaps, after you flip it,

you get **heads** with probability 3/4
and **tails** with probability 1/4. This is a probabilistic state of X.



https://izbicki.me/blog/how-to-create-an-unfair-coin-and-prove-it-with-math.html

A succinct way to represent this probabilistic state is by a **column vector**

$$\Pr(X = 0) = \tfrac{3}{4}$$
$$\Pr(X = 1) = \tfrac{1}{4}$$

$$\begin{pmatrix} 3/4 \\ 1/4 \end{pmatrix}$$

- All entries are **nonnegative real numbers**
- The sum of entries is 1

Representation of a vector as a linear combination

You can represent any vector in a given vector space as a unique linear combination of basis states.

For example, choose our basis to be $\{\vec{0}, \vec{1}\}$.

$$\vec{0} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \vec{1} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Then, the probabilistic state of the unfair coin can be represented as follows:

$$\Pr(X = 0) = \tfrac{3}{4}$$
$$\Pr(X = 1) = \tfrac{1}{4}$$

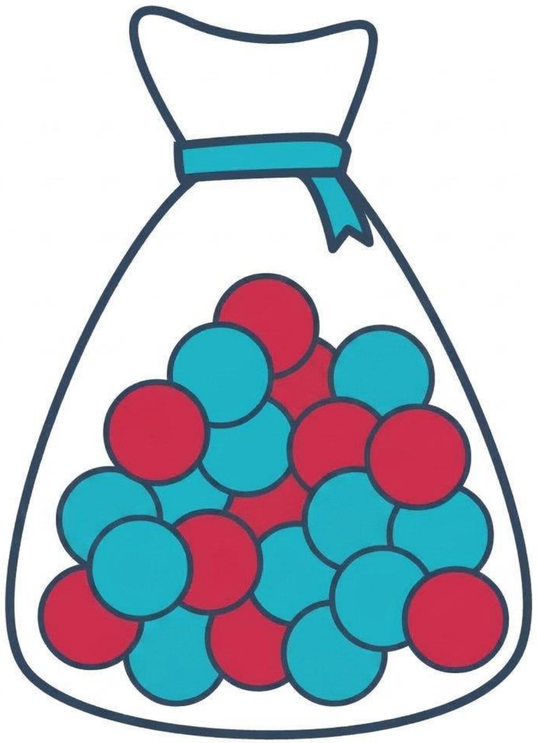
$$\begin{pmatrix} 3/4 \\ 1/4 \end{pmatrix} = \frac{3}{4}\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \frac{1}{4}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{3}{4}\vec{0} + \frac{1}{4}\vec{1}$$

Measuring probabilistic states

Consider the probabilistic state of a bag of 22 marbles

$$\Pr(X = red) = 10/22$$
$$\Pr(X = blue) = 12/22$$

$$\begin{pmatrix} 10/22 \\ 12/22 \end{pmatrix}$$



“Measuring” X reveals a transition, chosen at random, with certain probability

$$\begin{pmatrix} 10/22 \\ 12/22 \end{pmatrix} = \frac{10}{22}\vec{0} + \frac{12}{22}\vec{1}$$

$$\Pr = \frac{10}{22} \rightarrow \vec{0}$$

$$\Pr = \frac{12}{22} \rightarrow \vec{1}$$

Classical operations

Deterministic:
as a result of the operation, you get an exact result (no randomness!)

Probabilistic:
as a result of the operation, we may get different outcomes with certain probabilities

When representing classical states as vectors, it is natural to represent operations with matrices

For a vector with two entries, operator is a two-by-two matrix:

$$M = \begin{pmatrix} * & * \\ * & * \end{pmatrix}$$

(certain restriction apply to entry values)

Deterministic operations

$$\vec{0} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \vec{1} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Given any function $f : \Sigma \to \Sigma$, there is a (unique) matrix M that satisfies

$$\mathrm{M}|a\rangle = |f(a)\rangle \text{ for every } a \in \Sigma$$

- each column has exactly one 1
- all other entries are 0's

$\|a\rangle$	$\|f_2(a)\rangle$
$\vec{0}$	$\vec{0}$
$\vec{1}$	$\vec{1}$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad \therefore \quad M_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Deterministic operations

$$\vec{0} = \begin{pmatrix}1\\0\end{pmatrix}, \qquad \vec{1} = \begin{pmatrix}0\\1\end{pmatrix}$$

For example,

$\|a\rangle$	$\|f_1(a)\rangle$
$\vec{0}$	$\vec{0}$
$\vec{1}$	$\vec{0}$

$$\begin{pmatrix}1 & 1\\0 & 0\end{pmatrix}\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}1\\0\end{pmatrix}, \qquad \begin{pmatrix}1 & 1\\0 & 0\end{pmatrix}\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}1\\0\end{pmatrix} \quad \therefore \quad M_1 = \begin{pmatrix}1 & 1\\0 & 0\end{pmatrix}$$

$\|a\rangle$	$\|f_2(a)\rangle$
$\vec{0}$	$\vec{0}$
$\vec{1}$	$\vec{1}$

$$\begin{pmatrix}1 & 0\\0 & 1\end{pmatrix}\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}1\\0\end{pmatrix}, \qquad \begin{pmatrix}1 & 0\\0 & 1\end{pmatrix}\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}0\\1\end{pmatrix} \quad \therefore \quad M_2 = \begin{pmatrix}1 & 0\\0 & 1\end{pmatrix}$$

$\|a\rangle$	$\|f_3(a)\rangle$
$\vec{0}$	$\vec{1}$
$\vec{1}$	$\vec{0}$

$$\begin{pmatrix}0 & 1\\1 & 0\end{pmatrix}\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}0\\1\end{pmatrix}, \qquad \begin{pmatrix}0 & 1\\1 & 0\end{pmatrix}\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}1\\0\end{pmatrix} \quad \therefore \quad M_3 = \begin{pmatrix}0 & 1\\1 & 0\end{pmatrix}$$

$\|a\rangle$	$\|f_4(a)\rangle$
$\vec{0}$	$\vec{1}$
$\vec{1}$	$\vec{1}$

$$\begin{pmatrix}0 & 0\\1 & 1\end{pmatrix}\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}0\\1\end{pmatrix}, \qquad \begin{pmatrix}0 & 0\\1 & 1\end{pmatrix}\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}0\\1\end{pmatrix} \quad \therefore \quad M_4 = \begin{pmatrix}0 & 0\\1 & 1\end{pmatrix}$$

Probabilistic operations

$$\vec{0} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \vec{1} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

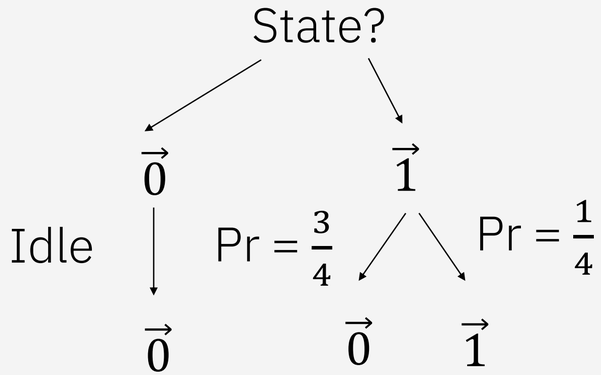
Probabilistic operations are classical operations that may introduce randomness or uncertainty, and are described by **stochastic** matrices:

- All entries are **nonnegative real numbers**
- Entries in each column add up to 1

For example,

a matrix describing the following stochastic (random) process:

If the classical state is 0, then do nothing.
If the classical state is 1, then flip the state with probability ¼.



$$\boldsymbol{M} = \begin{pmatrix} \mathbf{1} & \mathbf{1/4} \\ \mathbf{0} & \mathbf{3/4} \end{pmatrix} = \frac{1}{4}\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} + \frac{3}{4}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$M\vec{0} = \begin{pmatrix} 1 & 1/4 \\ 0 & 3/4 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \vec{0}$$

$$M\vec{1} = \begin{pmatrix} 1 & 1/4 \\ 0 & 3/4 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1/4 \\ 3/4 \end{pmatrix} = \frac{1}{4}\vec{0} + \frac{3}{4}\vec{1}$$

Dirac (bra-ket) notation in quantum

Classical → quantum: real to complex vector space.

In quantum mechanics/computing,

- $|\psi\rangle$ (ket) denotes the column vector in a complex vector space and physically represents a state of a quantum system
- $\langle\psi|$ (bra) denotes the row vector (transpose conjugate of ket)

$$|0\rangle = \begin{pmatrix}1\\0\end{pmatrix}, \qquad |1\rangle = \begin{pmatrix}0\\1\end{pmatrix}$$

$$|\psi\rangle = \begin{pmatrix}\alpha\\\beta\end{pmatrix} = \alpha|0\rangle + \beta|1\rangle$$

Transpose conjugate!
$\dagger\ is\ T^*$

$$\left(e^{i\theta}\right)^\dagger = \left(e^{-i\theta}\right)$$
$$(a+ib)^\dagger = a - ib$$

$$\langle 0| = |0\rangle^\dagger = (1\ 0), \quad \langle 1| = |1\rangle^\dagger = (0\ 1)$$

What makes such a state a valid quantum state?
Certain restrictions on entries!

Quantum information!

A quantum state of a system is represented by a column vector whose indices are placed in correspondence with the classical states of that system:

- The entries are **complex numbers**.
- The Euclidean norm must equal 1.

$$|\psi\rangle = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \dots \\ \alpha_n \end{pmatrix} \rightarrow |||\psi\rangle|| = \alpha_1^2 + \alpha_2^2 + \cdots + \alpha_n^2 = 1$$

Therefore, quantum state vectors are **unit vectors with respect to the norm**.

Example of qubit states

Standard (computational) basis: $\{|0\rangle, |1\rangle\}$

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Plus/minus states: $\{|+\rangle, |-\rangle\}$

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle, \qquad |-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$$

"Plus i" / "minus i" states: $\{|+i\rangle, |-i\rangle\}$

$$|+i\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle, \qquad |-i\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{i}{\sqrt{2}}|1\rangle$$

An arbitrary vector: $|\psi\rangle = \frac{1+2i}{3}|0\rangle - \frac{2}{3}|1\rangle$

Bra-ket notation exercise

Dagger == transpose conjugate

$$\langle\psi| = |\psi\rangle^\dagger = \left(\frac{1+2i}{3}|0\rangle - \frac{2}{3}|1\rangle\right)^\dagger = \frac{1-2i}{3}\langle 0| - \frac{2}{3}\langle 1|$$

Dirac (bra-ket) notation continued

Multiplying a row vector ($\langle a|$) by a column vector ($|b\rangle$) yields a scalar:

$$\langle a|b\rangle = \langle a||b\rangle = \begin{pmatrix} * & * & \dots & * \end{pmatrix} \begin{pmatrix} * \\ * \\ \dots \\ * \end{pmatrix} = (*) = *$$

$$\langle 0|0\rangle = \langle 0||0\rangle = \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$
$$= 1 * 1 + 0 * 0 = (1) = 1$$
$$\langle 1|0\rangle = \langle 1||0\rangle = \begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = (0) = 0$$

Multiplying a column vector ($|b\rangle$) by a row vector ($\langle a|$) yields a matrix:

$$|b\rangle\langle a| = \begin{pmatrix} * \\ * \\ \dots \\ * \end{pmatrix} \begin{pmatrix} * & * & \dots & * \end{pmatrix} = \begin{pmatrix} * & \cdots & * \\ \vdots & \ddots & \vdots \\ * & \cdots & * \end{pmatrix}$$

$$|0\rangle\langle 0| = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 * 1 & 1 * 0 \\ 0 * 1 & 0 * 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$|0\rangle\langle 1| = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$$|1\rangle\langle 0| = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

$$|1\rangle\langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

Re-arranging scalars/vectors/matrices in bra-ket notation



Quantum measurement

A measurement requires us to select a basis. For this explanation, let's focus on **the standard basis measurement**:

- The possible outcomes are the classical states.
- The probability for each classical state to be the outcome is the **absolute value squared** of the corresponding quantum state vector entry.

Standard (computational) basis:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Measuring the minus state $|-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$ yields

$$\Pr(|0\rangle) = |\frac{1}{\sqrt{2}}|^2 = \frac{1}{2}$$

$$\Pr(|1\rangle) = |-\frac{1}{\sqrt{2}}|^2 = \frac{1}{2}$$

Measuring the arbitrary state $|\psi\rangle = \frac{1+2i}{3}|0\rangle - \frac{2}{3}|1\rangle$ yields

$$\Pr(|0\rangle) = |\frac{1+2i}{3}|^2 = \frac{|1+2i|^2}{|3|^2} = \frac{\left|\sqrt{1+4}\right|^2}{9} = \frac{5}{9}$$

$$\Pr(|1\rangle) = |-\frac{2}{3}|^2 = \frac{4}{9}$$

Notice: $\frac{5}{9} + \frac{4}{9} = 1$

Bonus: measurement in an arbitrary basis

A measurement requires us to select a basis. Let's see how to measure in an arbitrary basis, using the bra-ket notation!

Standard (computational) basis:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Measuring an arbitrary state $|\psi\rangle$ in an arbitrary basis $\{|a\rangle, |b\rangle\}$

$$\Pr(|a\rangle) = |\langle a|\psi\rangle|^2$$
$$\Pr(|b\rangle) = |\langle b|\psi\rangle|^2$$

Measuring state $|-\rangle$ in the plus/minus basis $\{|+\rangle, |-\rangle\}$

$$\Pr(|+\rangle) = |0|^2 = 0$$
$$\Pr(|-\rangle) = |1|^2 = 1$$

Measuring state $|-\rangle$ in the computational basis

$$\Pr(|0\rangle) = \left|\langle 0| \left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle\right)\right|^2 = \frac{1}{2}$$

$$\Pr(|1\rangle) = \left|\langle 1| \left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle\right)\right|^2 = \frac{1}{2}$$

Bonus: measurement as an operator

You can also think about a measurements as projection operators.
In quantum mechanics, a **projection operator (or a projector)** is defined as $\mathbf{M}_n = |n\rangle\langle n|$ for all (normalized) $|n\rangle$ in the basis set.

Standard (computational) basis:

$$|0\rangle = \begin{pmatrix}1\\0\end{pmatrix}, \qquad |1\rangle = \begin{pmatrix}0\\1\end{pmatrix}$$

You can extract the portion of a state aligned with $|n\rangle$ if you apply a projector to a general quantum state $|\psi\rangle$:

$$M_n|\psi\rangle = |n\rangle\langle n|\psi\rangle = \langle n|\psi\rangle|n\rangle$$

Not normalized anymore!
Divide by its norm to re-normalize

Let's measure **expectation** of a projector M_n, $\langle M_n\rangle$.
For example, for $M_a = |a\rangle\langle a|$

$$\langle M_a\rangle = \langle\psi|M_a|\psi\rangle = \langle\psi|a\rangle\langle a|\psi\rangle = \langle a|\psi\rangle^*\langle a|\psi\rangle = \mathbf{|\langle a|\psi\rangle|^2 = Pr(|a\rangle)}$$

Single-qubit operations.
Superposition.



Unitary operations

Operations represented by unitary matrices – as a result of applying a unitary operation, the state (vector) remains a unit-vector

A square matrix U having complex number entries is unitary if it satisfies the following:

$$UU^\dagger = U^\dagger U = I$$

or, equivalently,

$$U^{-1} = U^\dagger$$

$$\begin{pmatrix} a & \cdots & b \\ \vdots & \ddots & \vdots \\ c & \cdots & d \end{pmatrix}^\dagger = \begin{pmatrix} a^* & \cdots & c^* \\ \vdots & \ddots & \vdots \\ b^* & \cdots & d^* \end{pmatrix}$$

Variety of unitary operations used in quantum computing

1. Pauli operations

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$X = \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$Y = \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\mathrm{Z} = \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Bit-flip:

$$X|0\rangle = |1\rangle$$
$$X|1\rangle = |0\rangle$$

Phase-flip:

$$Z|0\rangle = |0\rangle$$
$$Z|1\rangle = -|1\rangle$$

2. Hadamard

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$H|0\rangle = |+\rangle$$
$$H|1\rangle = |-\rangle$$

3. Phase

$$P_\theta = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix}$$

$$S = P_{\frac{\pi}{2}} = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$$

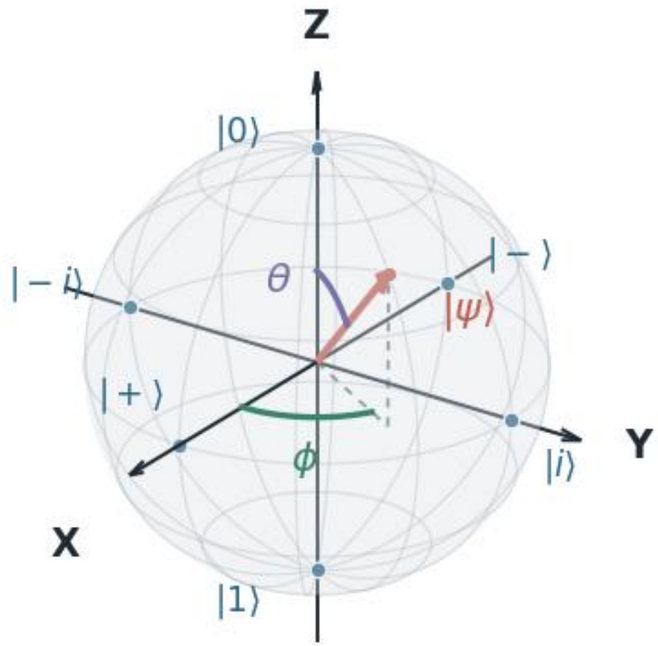
$$T = P_{\frac{\pi}{4}} = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1+i}{\sqrt{2}} \end{pmatrix}$$

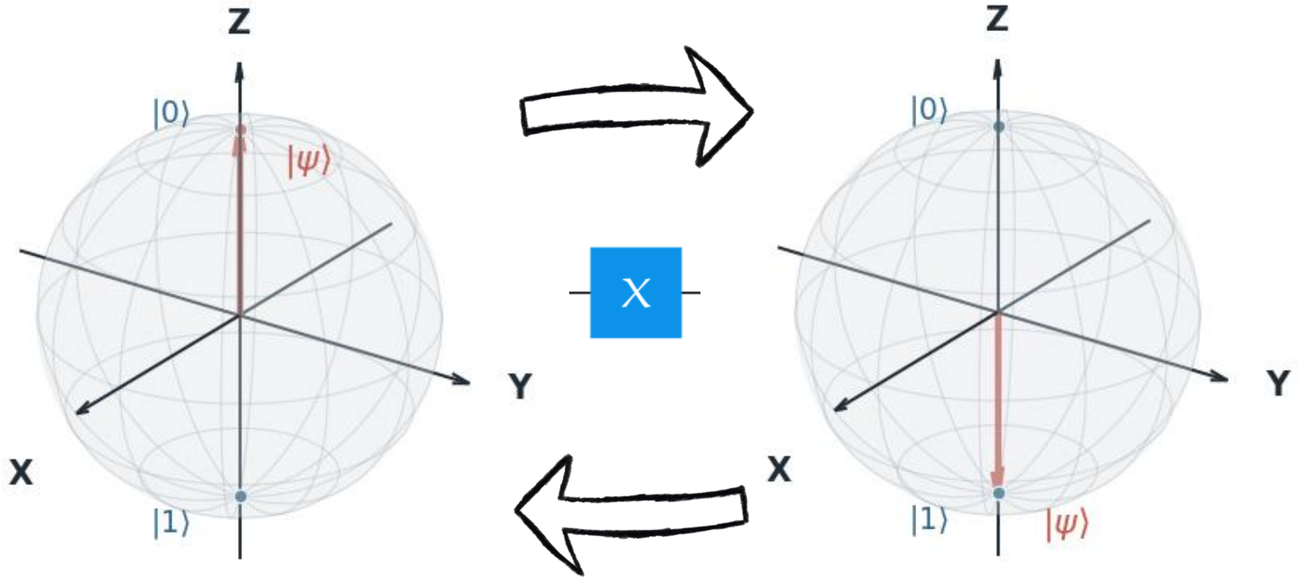
Bloch sphere representation

Polar angle $\theta = 45.0°$ Azimuthal angle $\phi = 60.0°$

Eigenstates Qubit state $|\psi\rangle$ Coordinate axes X, Y, Z

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$$





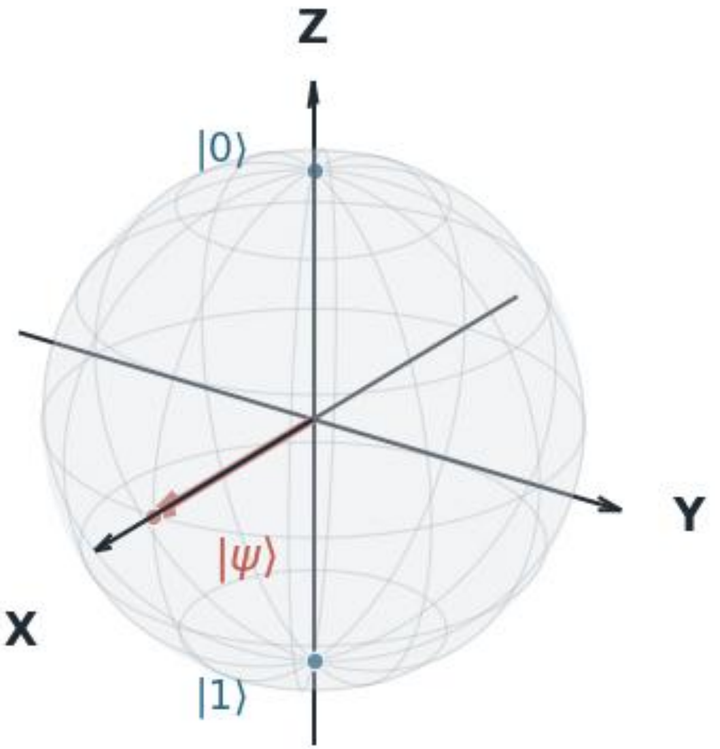
Superposition

You may have heard before a quantum mechanics term “superposition”.

What does it mean?

Quantum mechanics → Linear algebra: linear combination of basis states, for example:

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$



Do you recognize this state?

How would you represent state $|0\rangle$ as a superposition of $|+\rangle$ and $|-\rangle$?

NOTE: it is important to work in the right basis for your application

Multi-qubit operations.
Entanglement.



The Tensor Product

The **tensor product** of two vectors is another vector

$$|\phi\rangle = \sum_{a\in\Sigma} \alpha_a |a\rangle \quad and \quad |\psi\rangle = \sum_{b\in\Gamma} \beta_b |b\rangle$$

$$|\phi\rangle \otimes |\psi\rangle = \sum_{a\in\Sigma}\sum_{b\in\Gamma} \alpha_a \beta_b |ab\rangle$$

In general,

$$\begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_m \end{pmatrix} \otimes \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_k \end{pmatrix} = \begin{pmatrix} \alpha_1\beta_1 \\ \vdots \\ \alpha_1\beta_k \\ \alpha_2\beta_1 \\ \vdots \\ \alpha_2\beta_k \\ \vdots \\ \alpha_m\beta_1 \\ \vdots \\ \alpha_m\beta_k \end{pmatrix}$$

The Tensor Product

For example, consider the vectors

$$|\phi\rangle = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle = \frac{1}{2}\begin{pmatrix}1\\\sqrt{3}\end{pmatrix}$$

and

$$|\psi\rangle = \frac{2}{3}|0\rangle + \frac{\sqrt{5}}{3}|1\rangle = \frac{1}{3}\begin{pmatrix}2\\\sqrt{5}\end{pmatrix}$$

Their tensor product is written

$$|\phi\rangle\otimes|\psi\rangle = \frac{1}{2}|0\rangle \otimes \left(\frac{2}{3}|0\rangle + \frac{\sqrt{5}}{3}|1\rangle\right) + \frac{\sqrt{3}}{2}|1\rangle \otimes \left(\frac{2}{3}|0\rangle + \frac{\sqrt{5}}{3}|1\rangle\right)$$

$$= \frac{2}{6}|00\rangle + \frac{\sqrt{5}}{6}|01\rangle + \frac{\sqrt{3}}{6}|10\rangle + \frac{\sqrt{15}}{6}|11\rangle$$

Equivalently,

$$|\phi\rangle\otimes|\psi\rangle = \frac{1}{2}\begin{pmatrix}1\\\sqrt{3}\end{pmatrix} \otimes \frac{1}{3}\begin{pmatrix}2\\\sqrt{5}\end{pmatrix} = \frac{1}{6}\begin{pmatrix}1\begin{pmatrix}2\\\sqrt{5}\end{pmatrix}\\\sqrt{3}\begin{pmatrix}2\\\sqrt{5}\end{pmatrix}\end{pmatrix}$$

$$= \frac{1}{6}\begin{pmatrix}2\\\sqrt{5}\\\sqrt{3}\\\sqrt{15}\end{pmatrix}$$

The Tensor Product

The **tensor product** of two matrices is another matrix

$$\mathcal{M} = \sum_{a,b\in\Sigma} \alpha_{ab}|a\rangle\langle b| \quad and \quad \mathcal{N} = \sum_{c,d\in\Gamma} \beta_{cd}|c\rangle\langle d|$$

$$\mathcal{M}\otimes\mathcal{N} = \sum_{a,b\in\Sigma}\sum_{c,d\in\Gamma} \alpha_{ab}\beta_{cd}|ac\rangle\langle bd|$$

In general,

$$\begin{pmatrix} \alpha_{11} & \cdots & \alpha_{1m} \\ \vdots & \ddots & \vdots \\ \alpha_{m1} & \cdots & \alpha_{mm} \end{pmatrix} \otimes \begin{pmatrix} \beta_{11} & \cdots & \beta_{1k} \\ \vdots & \ddots & \vdots \\ \beta_{k1} & \cdots & \beta_{kk} \end{pmatrix} =$$

$$\begin{pmatrix} \alpha_{11}\beta_{11} & \cdots & \alpha_{11}\beta_{1k} & \cdots & \alpha_{1m}\beta_{11} & \cdots & \alpha_{1m}\beta_{1k} \\ \vdots & \ddots & \vdots & & \vdots & \ddots & \vdots \\ \alpha_{11}\beta_{k1} & \cdots & \alpha_{11}\beta_{kk} & & \alpha_{1m}\beta_{k1} & \cdots & \alpha_{1m}\beta_{kk} \\ & \vdots & & \ddots & & \vdots & \\ \alpha_{m1}\beta_{11} & \cdots & \alpha_{m1}\beta_{1k} & & \alpha_{mm}\beta_{11} & \cdots & \alpha_{mm}\beta_{1k} \\ \vdots & \ddots & \vdots & \cdots & \vdots & \ddots & \vdots \\ \alpha_{m1}\beta_{k1} & \cdots & \alpha_{m1}\beta_{kk} & & \alpha_{mm}\beta_{k1} & \cdots & \alpha_{mm}\beta_{kk} \end{pmatrix}$$

The Tensor Product

For example,

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \otimes \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} =$$

$$\begin{pmatrix} 1\begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} & 2\begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} \\ 3\begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} & 4\begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} \end{pmatrix} =$$

$$\begin{pmatrix} 5 & 6 & 10 & 12 \\ 7 & 8 & 14 & 16 \\ 15 & 18 & 20 & 24 \\ 21 & 24 & 28 & 32 \end{pmatrix}$$



Quantum states in joint systems

Quantum state vectors of the pair (X,Y) may be expressed in many equivalent forms:

$$|\psi\rangle = \frac{1}{2}|0\rangle|0\rangle + \frac{i}{2}|0\rangle|1\rangle - \frac{1}{2}|1\rangle|0\rangle - \frac{i}{2}|1\rangle|1\rangle$$

$$|\psi\rangle = \frac{1}{2}|00\rangle + \frac{i}{2}|01\rangle - \frac{1}{2}|10\rangle - \frac{i}{2}|11\rangle$$

$$|\psi\rangle = \frac{1}{2}\begin{pmatrix} 1 \\ i \\ -1 \\ -i \end{pmatrix}$$

$$|\psi\rangle = \frac{1}{2}|0\rangle \otimes |0\rangle + \frac{i}{2}|0\rangle \otimes |1\rangle - \frac{1}{2}|1\rangle \otimes |0\rangle - \frac{i}{2}|1\rangle \otimes |1\rangle$$

$$|\psi\rangle = \frac{1}{2}|0\rangle_X|0\rangle_Y + \frac{i}{2}|0\rangle_X|1\rangle_Y - \frac{1}{2}|1\rangle_X|0\rangle_Y - \frac{i}{2}|1\rangle_X|1\rangle_Y$$

Quantum states that can be written as the tensor product between two other quantum states are called **product states**
For example,

$$|\psi\rangle = \frac{1}{2}|00\rangle + \frac{i}{2}|01\rangle - \frac{1}{2}|10\rangle - \frac{i}{2}|11\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle) = |\phi\rangle_1 \otimes |\phi\rangle_2$$

On the other hand, a quantum state such as

$$|\psi\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

is **not** a product state, as it cannot be written as a tensor product.

Entangled states

Consider this quantum state

$$|\psi\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\0\\0\\1\end{pmatrix}$$

Famous quantum states

The previous example is one of the four **Bell states,** which collectively form the **Bell basis**

$$|\phi^+\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle \qquad |\psi^+\rangle = \frac{1}{\sqrt{2}}|01\rangle + \frac{1}{\sqrt{2}}|10\rangle$$

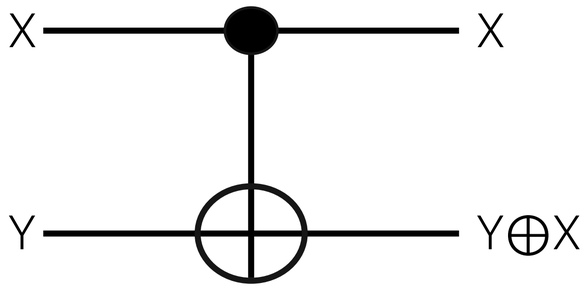
$$|\phi^-\rangle = \frac{1}{\sqrt{2}}|00\rangle - \frac{1}{\sqrt{2}}|11\rangle \qquad |\psi^-\rangle = \frac{1}{\sqrt{2}}|01\rangle - \frac{1}{\sqrt{2}}|10\rangle$$

For three-qubit states, there is the well-known **GHZ state** and the **W state**

$$|GHZ\rangle = \frac{1}{\sqrt{2}}|000\rangle + \frac{1}{\sqrt{2}}|111\rangle \qquad |W\rangle = \frac{1}{\sqrt{3}}|001\rangle + \frac{1}{\sqrt{3}}|010\rangle\frac{1}{\sqrt{3}}|100\rangle$$

Quantum Logic Gates -- CNOT

Operator	Gate(s)		Matrix
Pauli-X (X)	X	⊕	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
Pauli-Y (Y)	Y		$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$
Pauli-Z (Z)	Z		$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
Hadamard (H)	H		$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
Phase (S, P)	S		$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$
π/8 (T)	T		$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$
Controlled Not (CNOT, CX)			$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$
Controlled Z (CZ)	Z		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$
SWAP			$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Toffoli (CCNOT, CCX, TOFF)			$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$



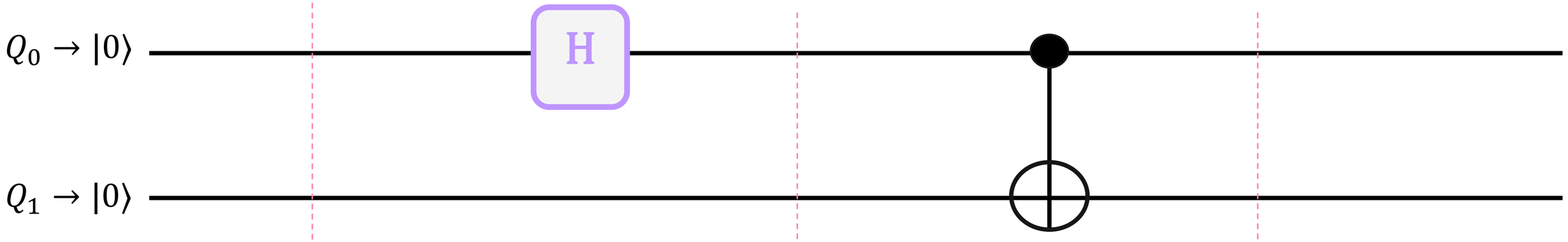
A **controlled-NOT** operation on two bits X and Y is described by

- If X = 1, then perform a NOT operation on Y
- Otherwise, do nothing.

X is the **control** bit that determines whether or not a NOT operation is applied to the **target** bit Y.

Input X	Input Y	Output X	Output Y+X
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0

Generating an Entangled State



$$|\psi\rangle = |0\rangle \otimes |0\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$(H \otimes I)|00\rangle$$

$$= \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle$$

$$= \frac{|00\rangle + |10\rangle}{\sqrt{2}}$$

$$CNOT\left(\frac{|00\rangle + |10\rangle}{\sqrt{2}}\right)$$

$$= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 1/\sqrt{2} \\ 0 \\ 0 \\ 1/\sqrt{2} \end{pmatrix}$$

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \frac{|00\rangle + |11\rangle}{\sqrt{2}} = |\phi^+\rangle$$

Quantum Measurements

A quantum state vector of (X,Y) takes the form

$$|\psi\rangle = \sum_{a\in\Sigma}\sum_{b\in\Gamma} \alpha_{ab}|ab\rangle$$

If both subsystems are measured, then each outcome occurs with probability

$$|\langle ab|\psi\rangle|^2 = |\alpha_{ab}|^2$$

If just X is measured, however, the probability for each outcome is

$$\Pr(outcome\ is\ a) = \sum_{b\in\Gamma} |\langle ab|\psi\rangle|^2 = \sum_{b\in\Gamma} |\alpha_{ab}|^2$$

Quantum Measurements

For example, suppose the (X,Y) system is in the state

$$|\psi\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{2}|01\rangle + \frac{i}{2\sqrt{2}}|10\rangle - \frac{1}{2\sqrt{2}}|11\rangle$$

$$= |0\rangle \otimes (\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{2}|1\rangle) + |1\rangle \otimes (\frac{i}{2\sqrt{2}}|0\rangle - \frac{1}{2\sqrt{2}}|1\rangle)$$

The probability that we measure X and yield 0 is

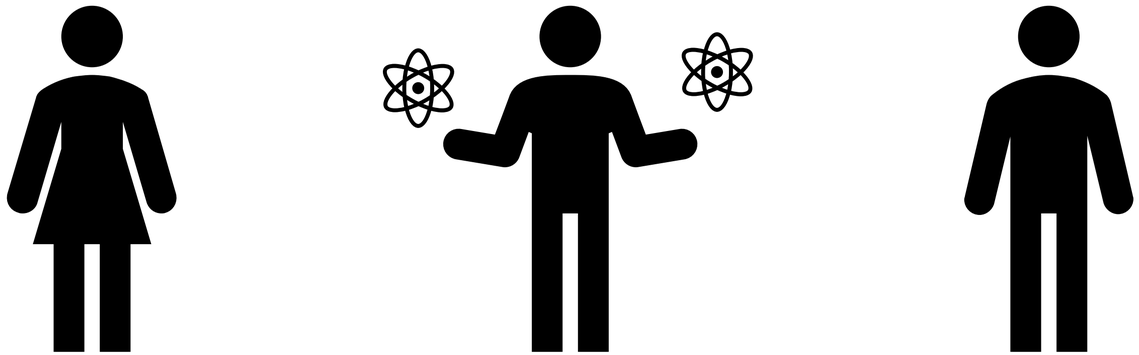
$$P(X=0) = \|(\langle 0|_X \otimes \mathbb{I}_Y)|\psi\rangle\|^2$$
$$= (\frac{1}{\sqrt{2}}\langle 0| + \frac{1}{2}\langle 1|)(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{2}|1\rangle)$$
$$= \frac{1}{2}\langle 0|0\rangle + \frac{1}{2\sqrt{2}}\langle 0|1\rangle + \frac{1}{2\sqrt{2}}\langle 1|0\rangle + \frac{1}{4}\langle 1|1\rangle$$
$$= \frac{1}{2} + 0 + 0 + \frac{1}{4}$$
$$= \frac{3}{4}$$

Consequently, this **collapses** the Y-subsystem to the state

$$|Y\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{2}|1\rangle \longrightarrow \sqrt{\frac{2}{3}}|0\rangle + \frac{1}{\sqrt{3}}|1\rangle$$

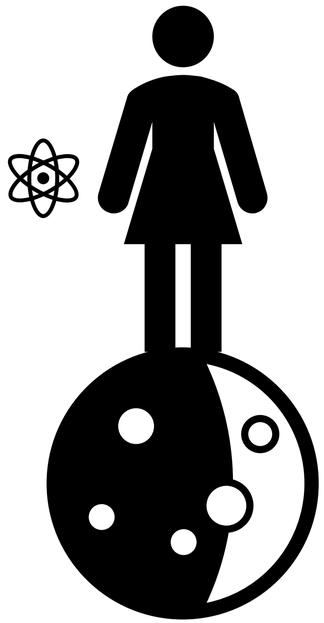
What does it mean to be entangled? Bell's inequality

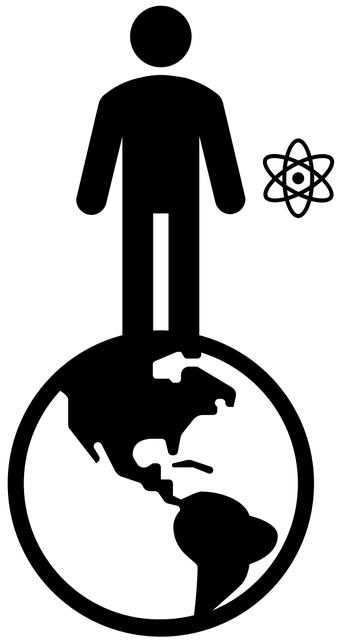
Consider the following thought experiment:



What does it mean to be entangled? Bell's inequality

Consider the following thought experiment:

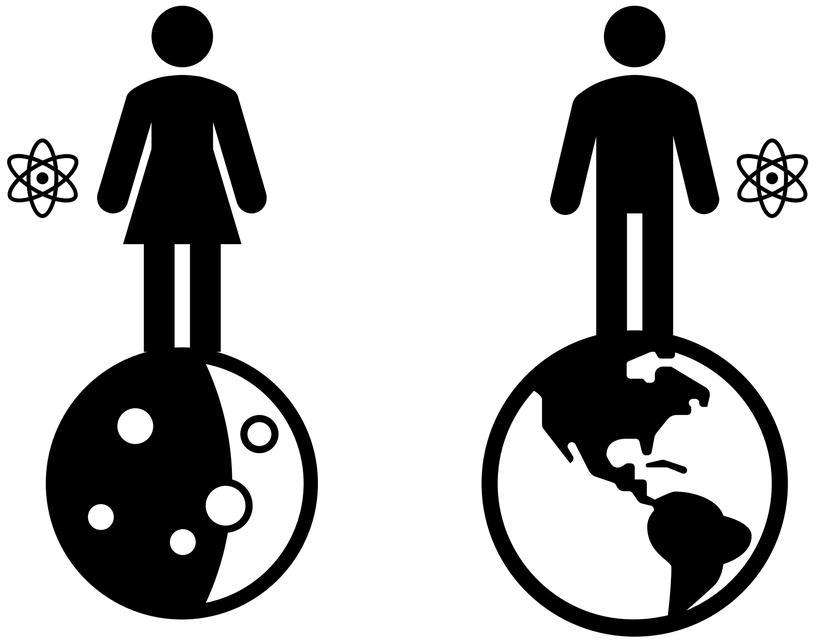




What does it mean to be entangled? Bell's inequality

Consider the following thought experiment:

$$A_0, A_1$$
$$\langle A_0 \rangle \equiv a_0, \langle A_1 \rangle \equiv a_1$$
$$\implies a_0, a_1 = \pm 1$$



$$B_0, B_1$$
$$\langle B_0 \rangle \equiv b_0, \langle B_1 \rangle \equiv b_1$$
$$\implies b_0, b_1 = \pm 1$$

$$a_0 b_0 + a_0 b_1 + a_1 b_0 - a_1 b_1$$
$$= (a_0 + a_1) b_0 + (a_0 - a_1) b_1$$

$$|\langle A_0 B_0 \rangle + \langle A_0 B_1 \rangle + \langle A_1 B_0 \rangle - \langle A_1 B_1 \rangle| \leq 2$$

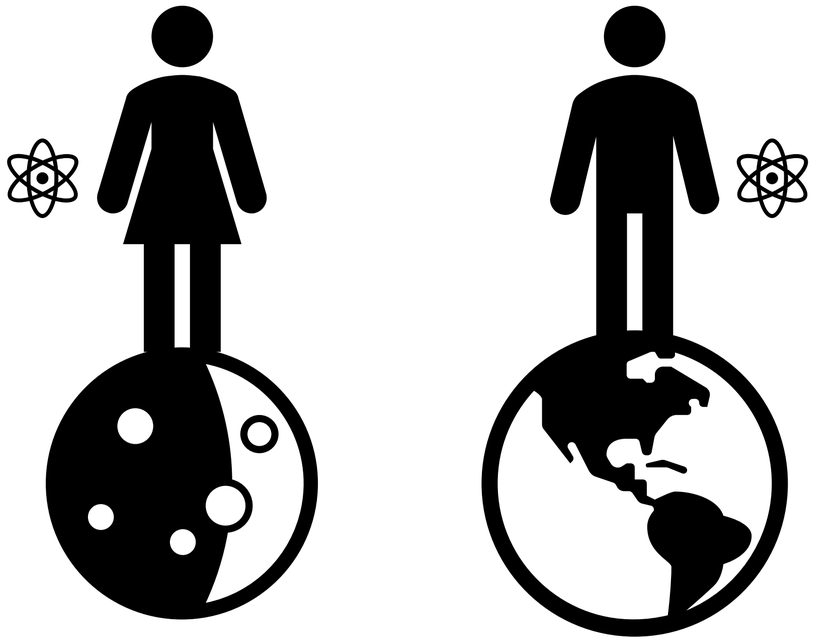
What does it mean to be entangled? Bell's inequality

Consider the following thought experiment:

$$|\psi^-\rangle = \frac{1}{\sqrt{2}}|01\rangle - \frac{1}{\sqrt{2}}|10\rangle$$

$$A_0 \equiv \sigma_z \, ; A_1 \equiv \sigma_x$$

$$B_0 \equiv -\frac{\sigma_x + \sigma_z}{\sqrt{2}} \, ; B_1 \equiv \frac{\sigma_x - \sigma_z}{\sqrt{2}}$$



$$|\langle A_0 B_0 \rangle + \langle A_0 B_1 \rangle + \langle A_1 B_0 \rangle - \langle A_1 B_1 \rangle| \leq 2$$

What does it mean to be entangled? Bell's inequality

Consider the following thought experiment:

$$|\psi^-\rangle = \frac{1}{\sqrt{2}}|01\rangle - \frac{1}{\sqrt{2}}|10\rangle$$

$$A_0 \equiv \sigma_z \; ; A_1 \equiv \sigma_x$$

$$B_0 \equiv -\frac{\sigma_x + \sigma_z}{\sqrt{2}} ; B_1 \equiv \frac{\sigma_x - \sigma_z}{\sqrt{2}}$$

$$A_0 \otimes B_0 = \sigma_z \otimes -\frac{1}{\sqrt{2}}(\sigma_x + \sigma_z) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \otimes -\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$= -\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

$$\langle\psi^-|(A_0 \otimes B_0)|\psi^-\rangle \equiv \langle A_0 B_0\rangle$$

$$= -\frac{1}{2\sqrt{2}}\begin{pmatrix} 0 & 1 & -1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix}$$

$$= -\frac{1}{2\sqrt{2}}\begin{pmatrix} 0 & 1 & -1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ -1 \\ 1 \\ 1 \end{pmatrix}$$

$$= \frac{1}{\sqrt{2}}$$

$$|\langle A_0 B_0\rangle + \langle A_0 B_1\rangle + \langle A_1 B_0\rangle - \langle A_1 B_1\rangle| \leq 2$$

What does it mean to be entangled? Bell's inequality

Consider the following thought experiment:

$$|\psi^-\rangle = \frac{1}{\sqrt{2}}|01\rangle - \frac{1}{\sqrt{2}}|10\rangle$$

$$A_0 \equiv \sigma_z \, ; A_1 \equiv \sigma_x$$

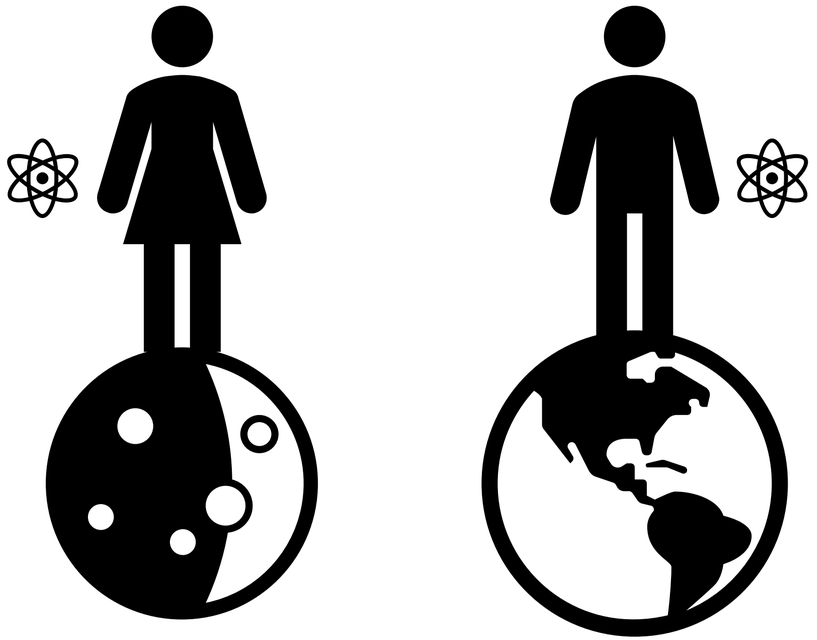
$$B_0 \equiv -\frac{\sigma_x + \sigma_z}{\sqrt{2}} \, ; B_1 \equiv \frac{\sigma_x - \sigma_z}{\sqrt{2}}$$

$$\langle A_0 B_0 \rangle = \frac{1}{\sqrt{2}}$$

$$\langle A_0 B_1 \rangle = \frac{1}{\sqrt{2}}$$

$$\langle A_1 B_0 \rangle = \frac{1}{\sqrt{2}}$$

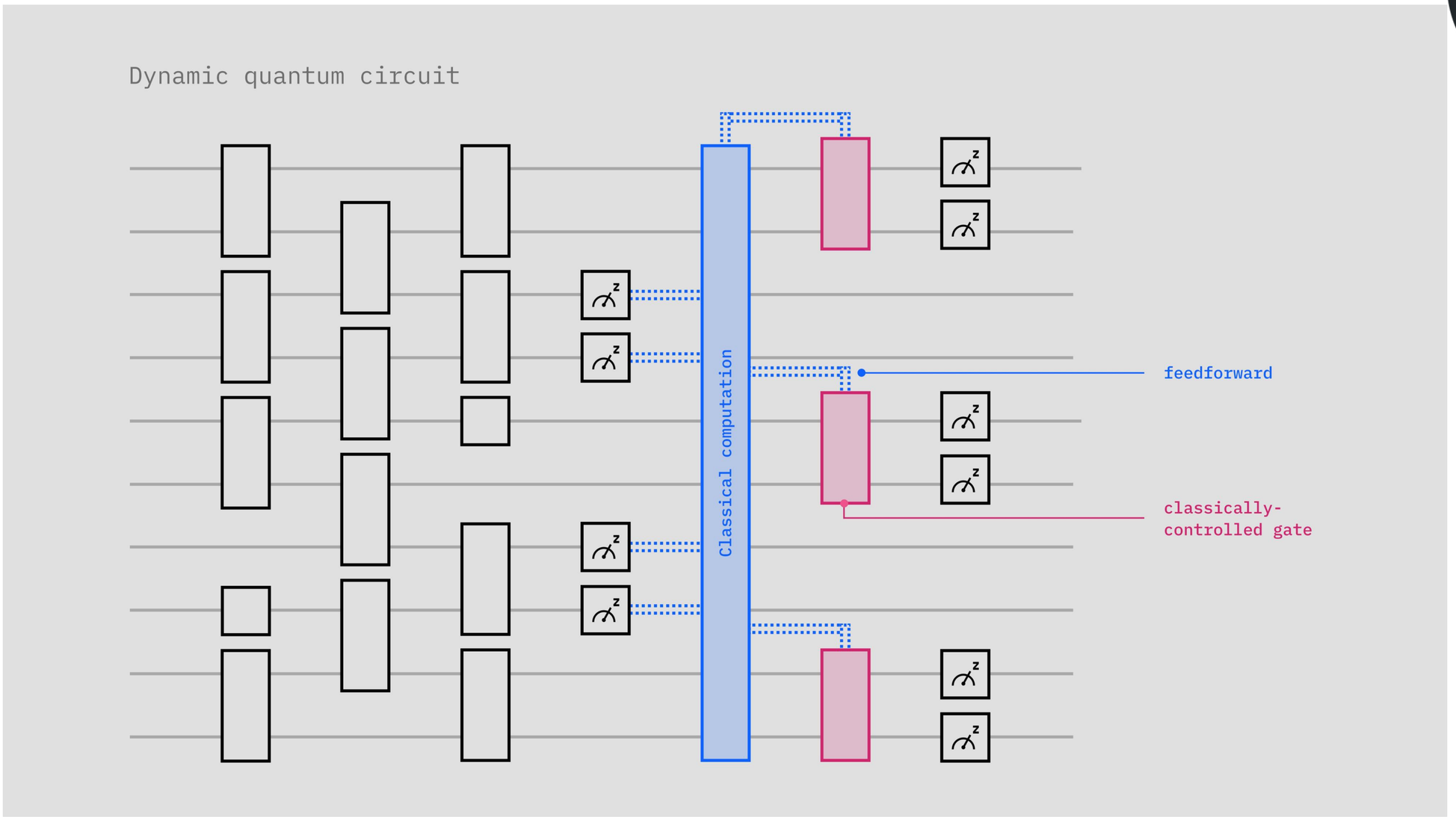
$$\langle A_1 B_1 \rangle = -\frac{1}{\sqrt{2}}$$



$$|\langle A_0 B_0 \rangle + \langle A_0 B_1 \rangle + \langle A_1 B_0 \rangle - \langle A_1 B_1 \rangle| \leq 2$$

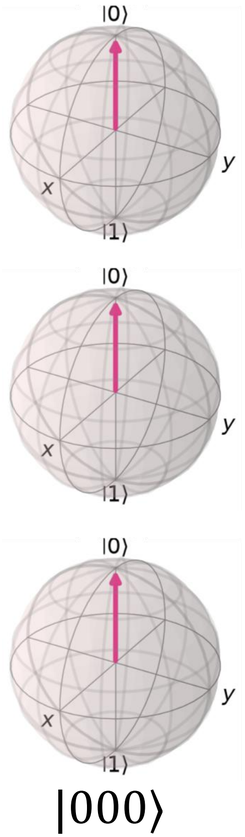
$$\left|2\sqrt{2}\right| \leq 2$$

Quantum Circuits



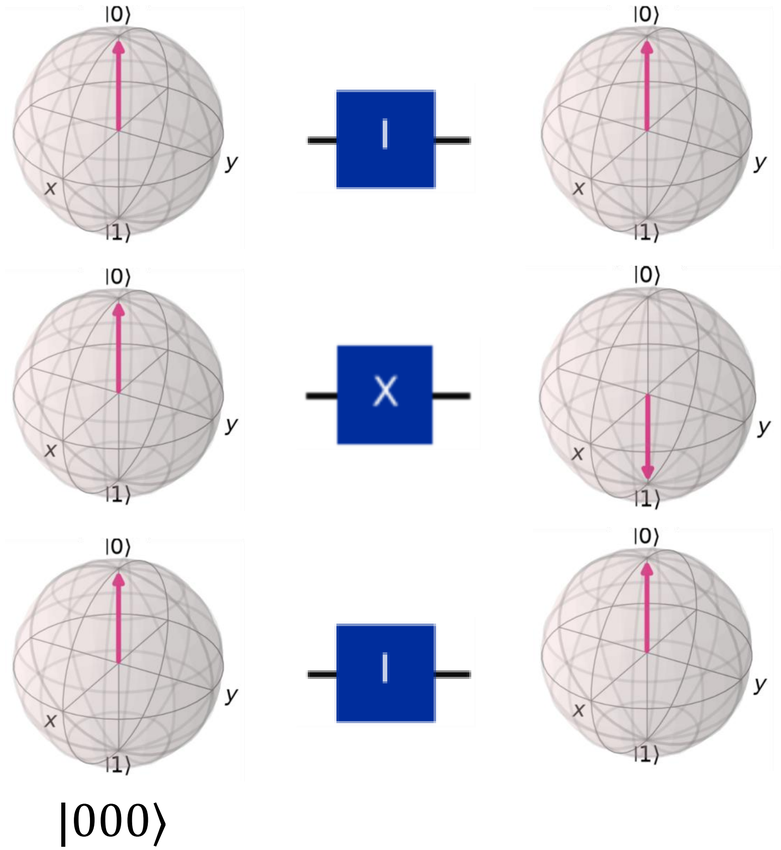
Building a quantum circuit

- Qubit initialization
- State preparation
- Gate operations of interest
- Measurement



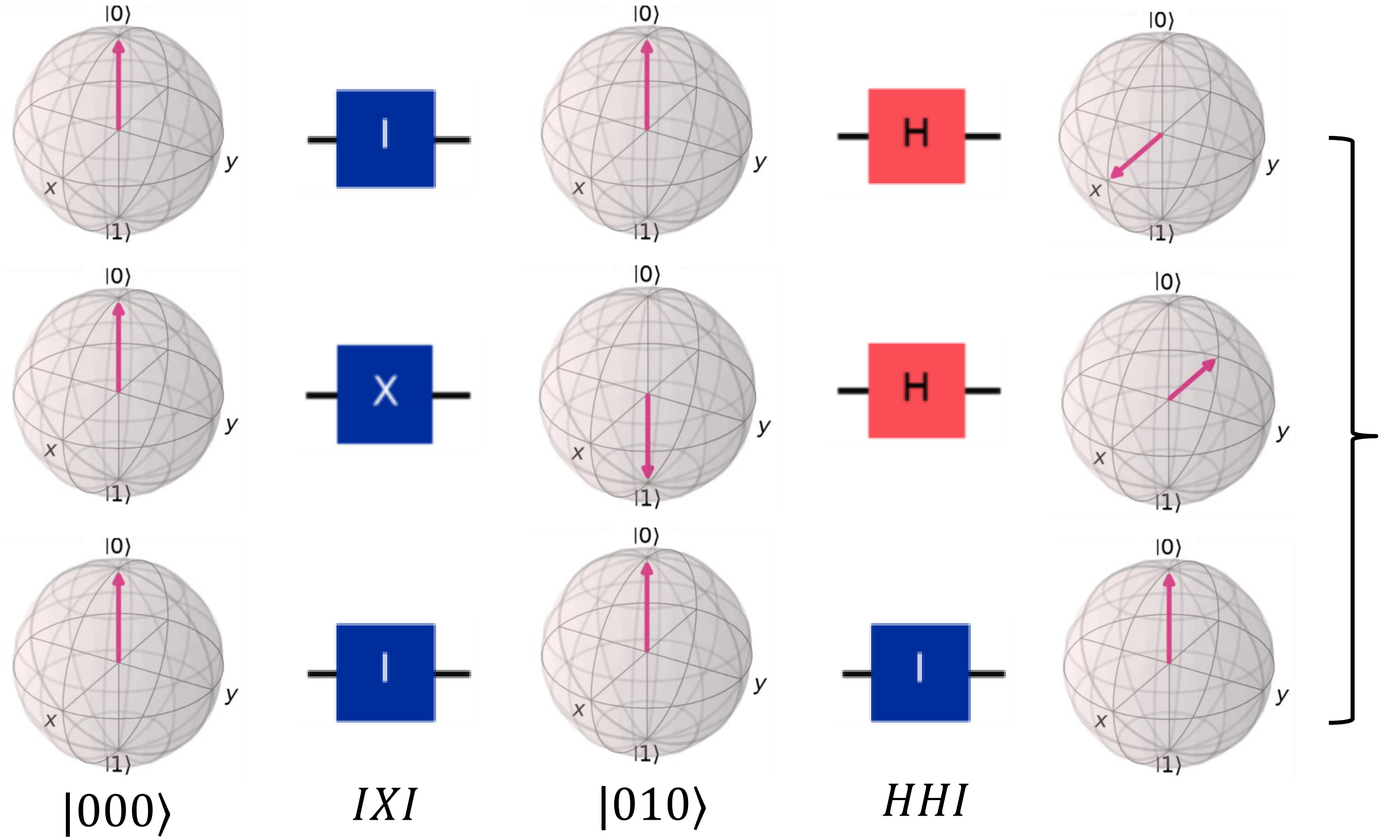
Building a quantum circuit

- Qubit initialization
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Building a quantum circuit

- Qubit initialization
- State preparation
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- Measurement

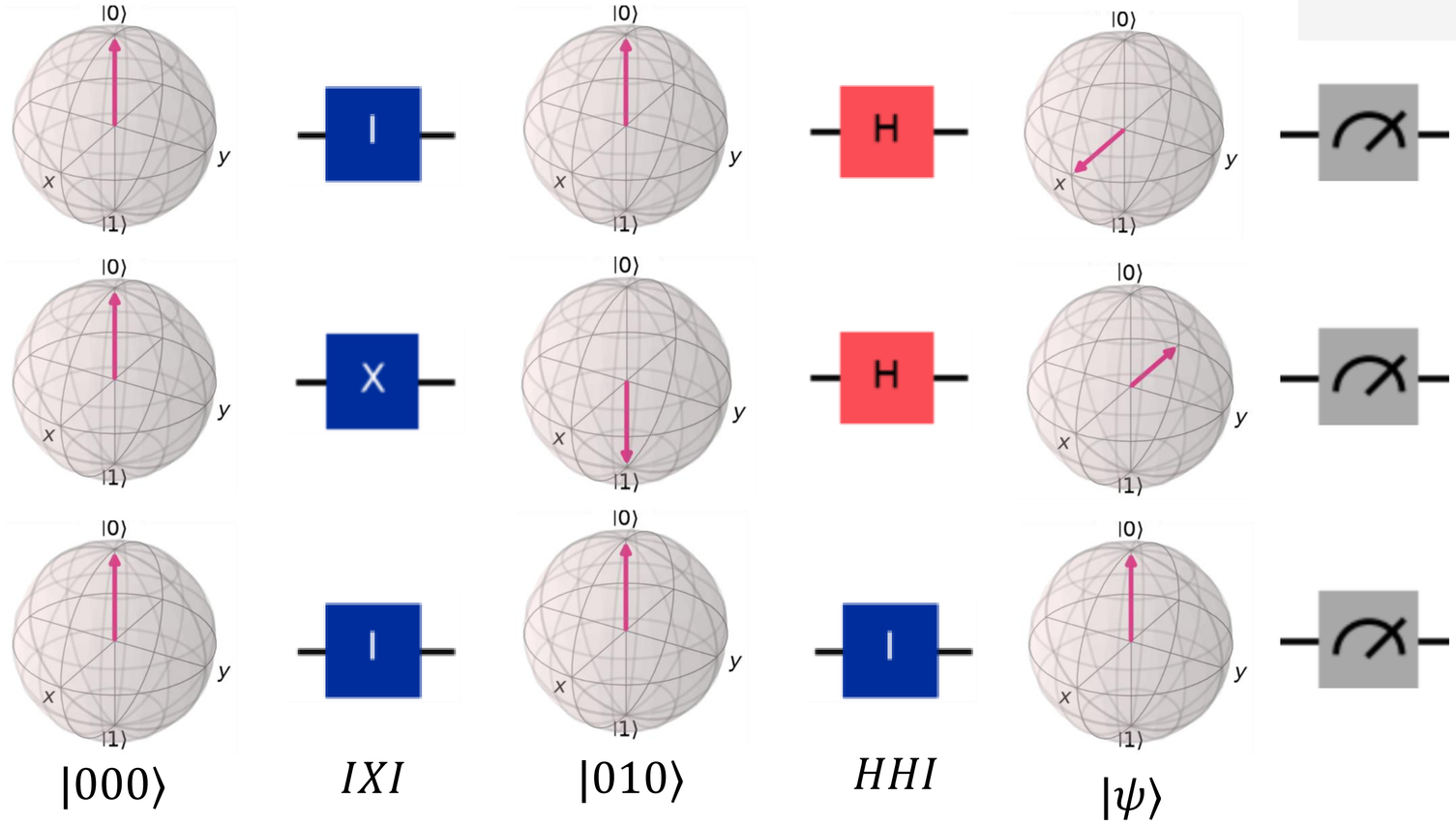


$$|\psi\rangle = \frac{1}{2}\begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

Building a quantum circuit

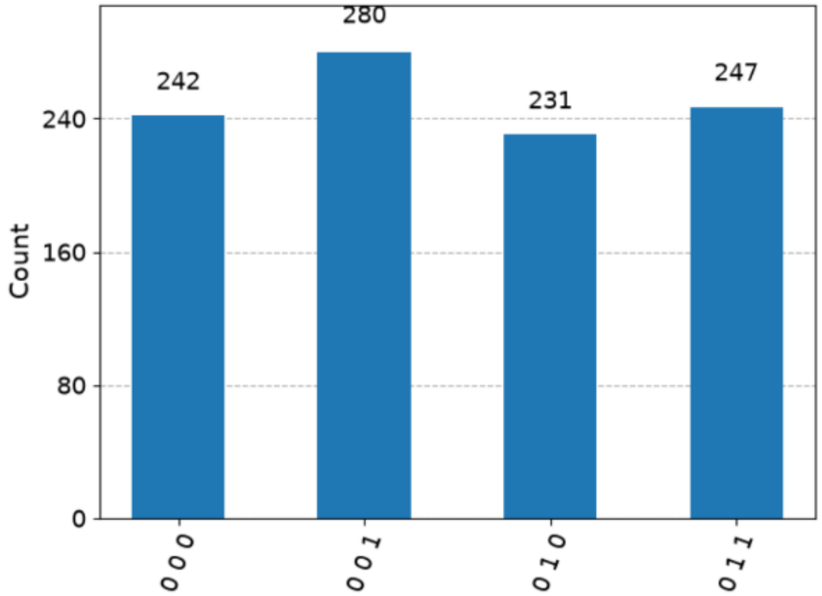
- Qubit initialization
- State preparation
- Gate operations of interest
- Measurement

$$|\psi\rangle = \frac{1}{2}\begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad C = \begin{Bmatrix} |000\rangle, |001\rangle, |010\rangle, |011\rangle, \\ |100\rangle, |101\rangle, |110\rangle, |111\rangle \end{Bmatrix}$$



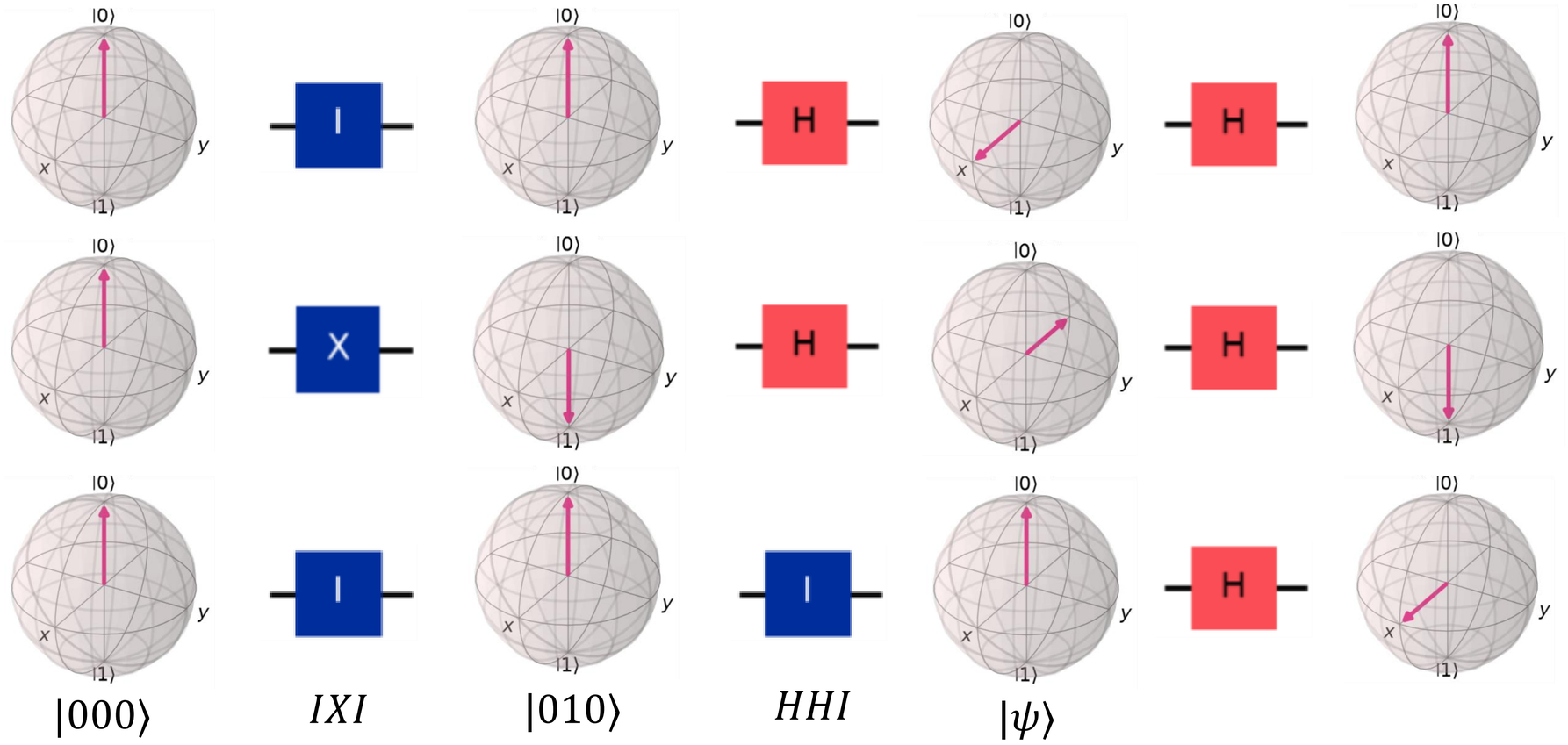
$|000\rangle$ IXI $|010\rangle$ HHI $|\psi\rangle$

Measurement : $\mathrm{P}(c_i) = |\langle c_i|\psi\rangle|^2$



Switching measurement basis before measurement

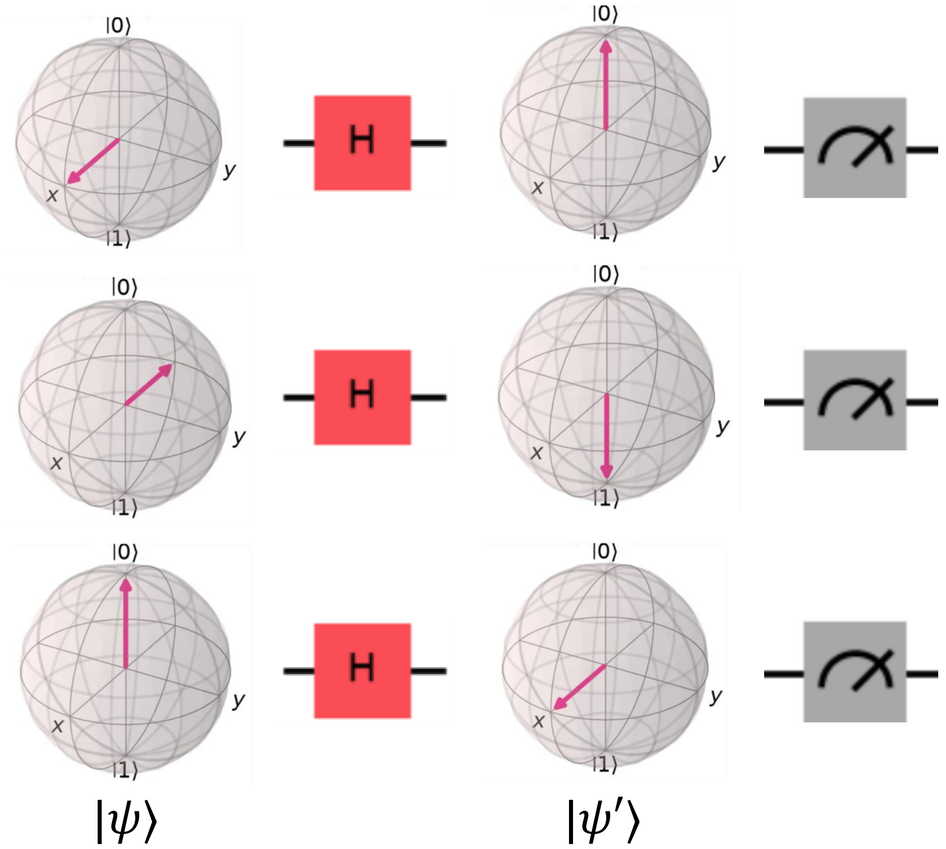
- Hadamard gate H transforms qubit state from Z to X basis and vice versa



$$|\psi'\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}0\\0\\1\\1\\0\\0\\0\\0\end{pmatrix}$$

Switching measurement basis before measurement

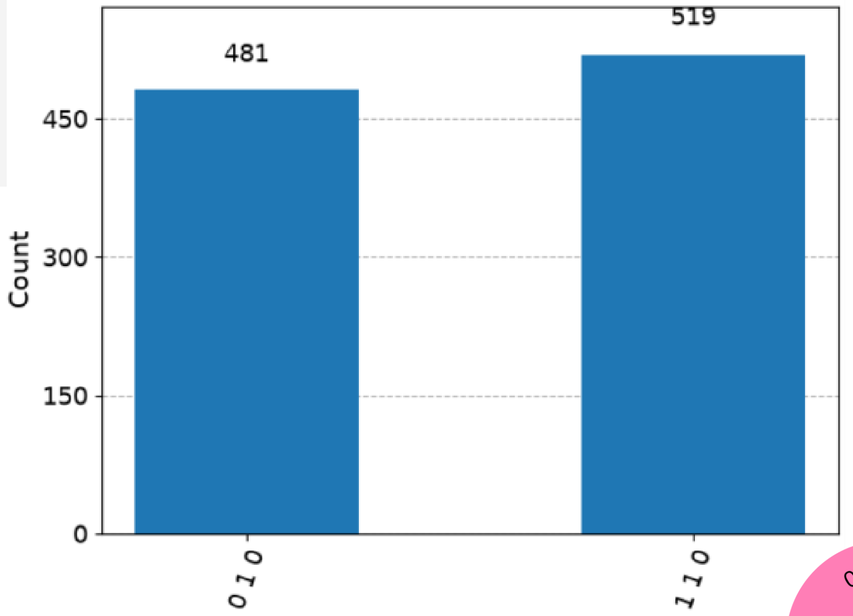
- Hadamard gate H transforms qubit state from Z to X basis and vice versa



$$|\psi'\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$C = \begin{Bmatrix} |000\rangle, |001\rangle, |010\rangle, |011\rangle, \\ |100\rangle, |101\rangle, |110\rangle, |111\rangle \end{Bmatrix}$$

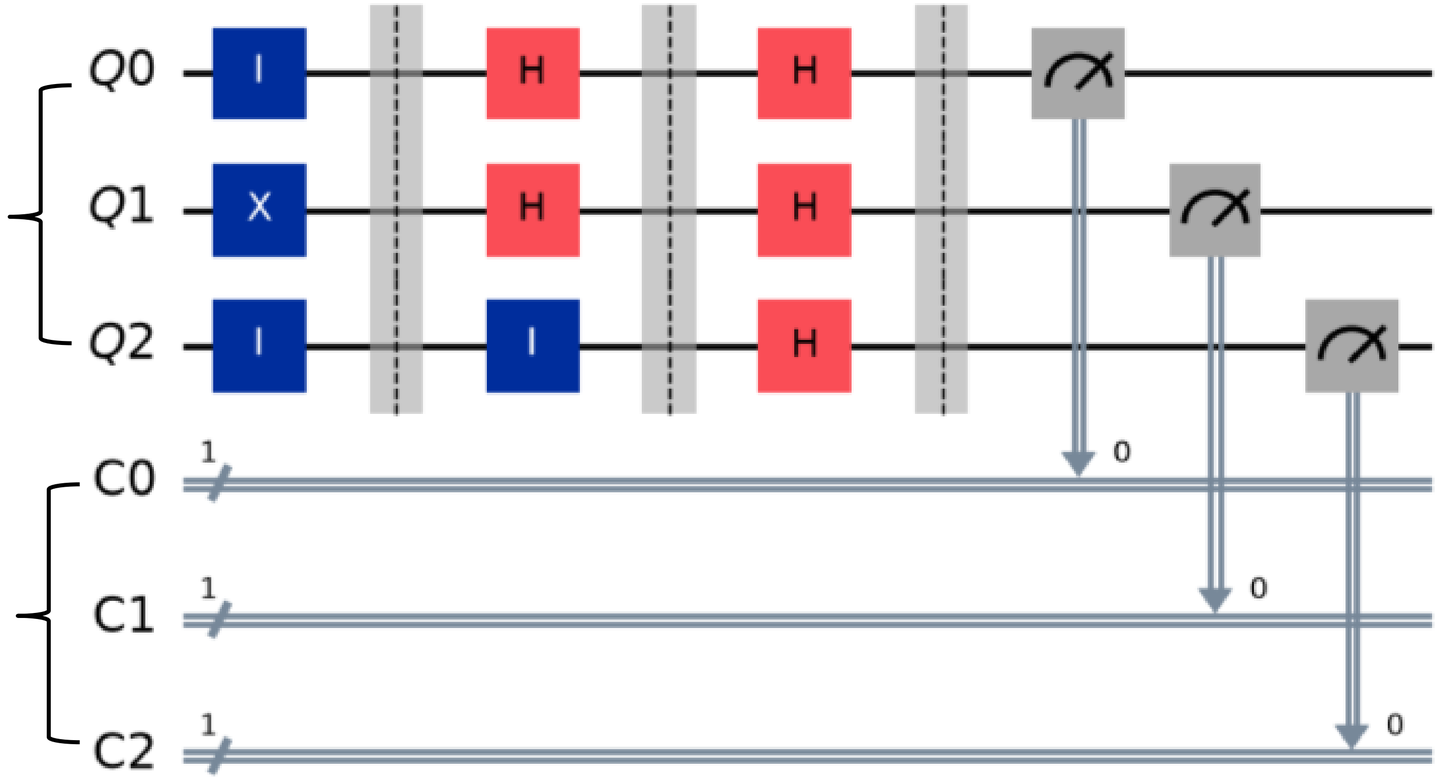
Measurement : $\mathrm{P}(c_i) = |\langle c_i|\psi'\rangle|^2$



Defining quantum circuits

Quantum registers – collection of qubits

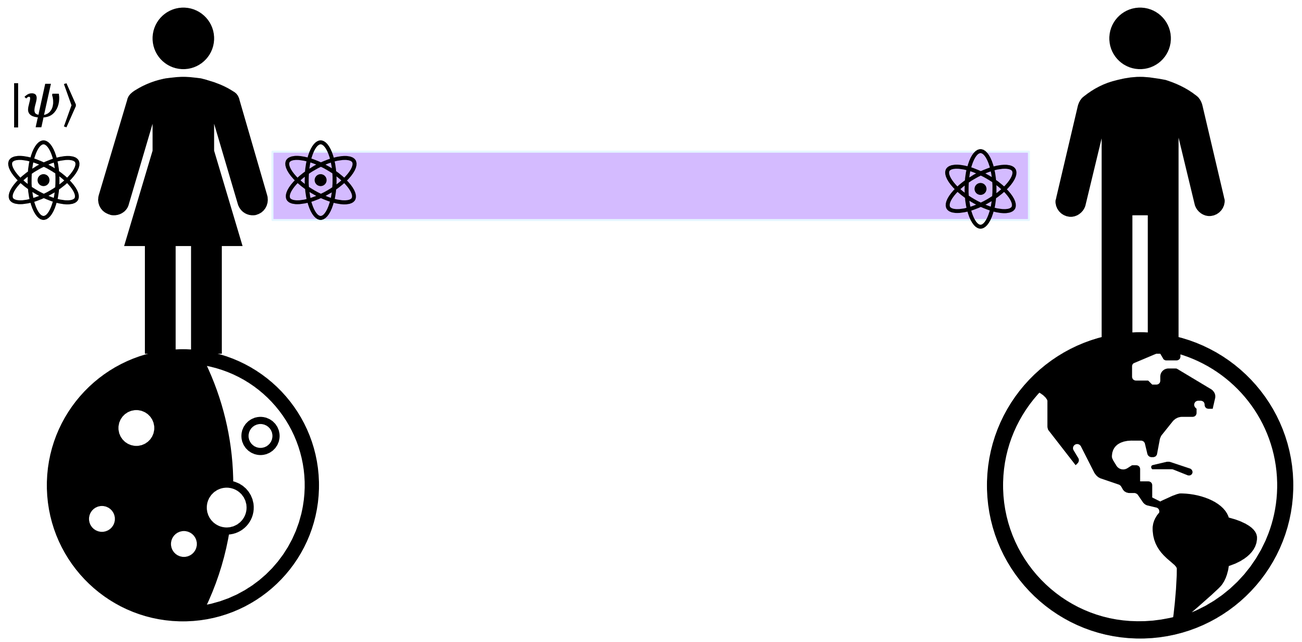
Classical registers – store final measurement results as classical bits



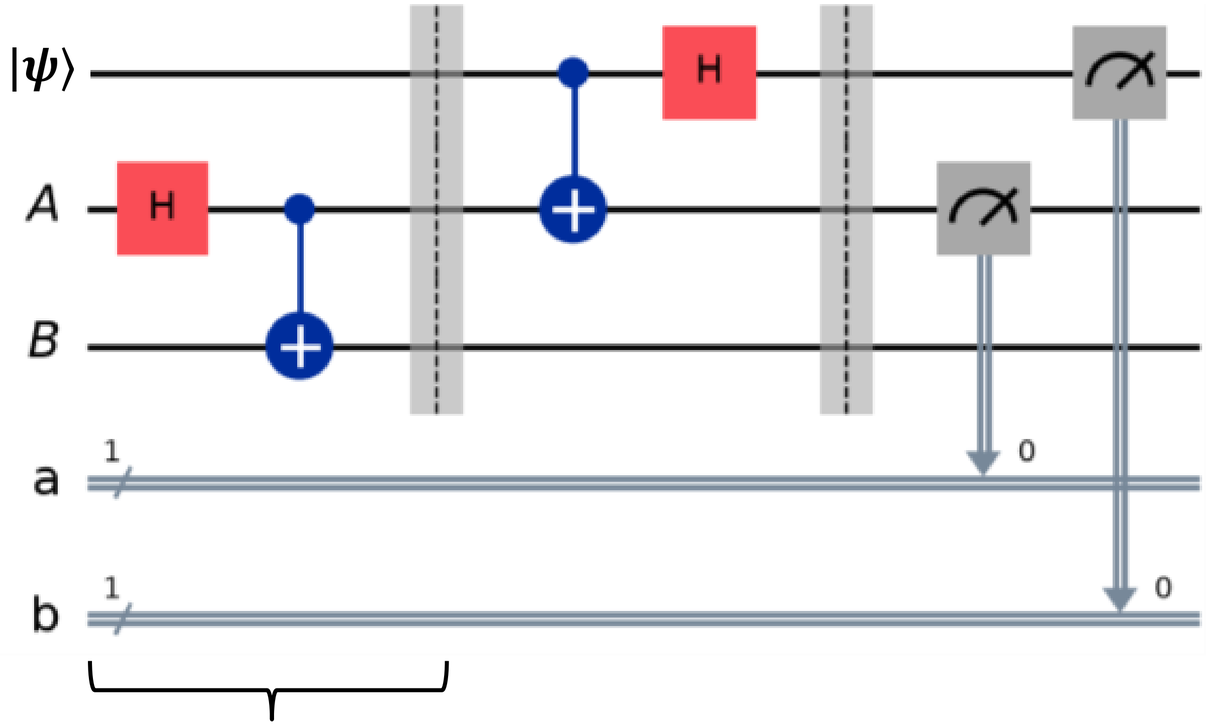
*** Keep in mind that the Qiskit qubit ordering is **reversed** from the quantum circuit visualization!
While Q0 is displayed as the topmost qubit and Q2 as the bottom qubit, the state representation is reversed: $|Q2\rangle \otimes |Q1\rangle \otimes |Q0\rangle$. ***

Example walkthrough: Quantum teleportation

Alice wants to transmit her qubit state $|\psi\rangle$ to Bob by making use of a shared entangled quantum state



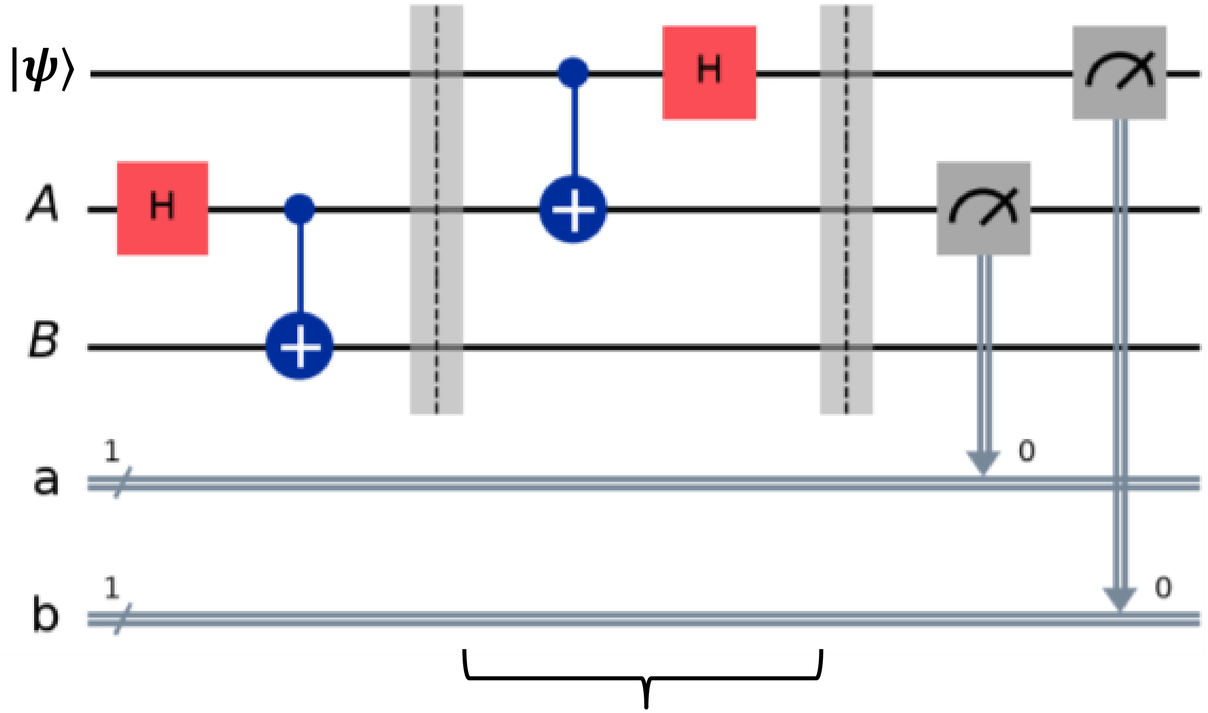
Example walkthrough: Quantum teleportation



Alice and Bob generate an entangled Bell state

$$|\phi^+\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\0\\0\\1\end{pmatrix} = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \longrightarrow |\psi\rangle \otimes |\phi^+\rangle$$

Example walkthrough: Quantum teleportation



Let's assume an arbitrary state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

$$(H \otimes I)\ \mathrm{CNOT}\ (|\psi\rangle \otimes |\phi^+\rangle) = \frac{\alpha|+\rangle(|00\rangle + |11\rangle) + \beta|-\rangle(|00\rangle + |11\rangle)}{\sqrt{2}}$$

$$= \frac{1}{2}[(\alpha|0\rangle + \beta|1\rangle)|00\rangle + (\alpha|0\rangle - \beta|1\rangle)|01\rangle + (\alpha|1\rangle + \beta|0\rangle)|10\rangle + (\alpha|1\rangle - \beta|0\rangle)|11\rangle]$$

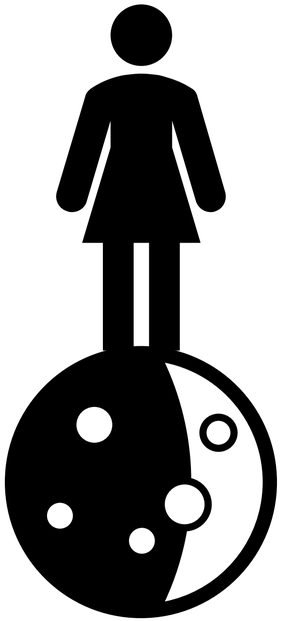
Example walkthrough: Quantum teleportation

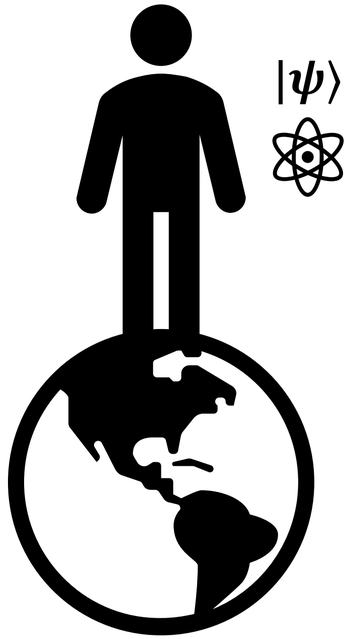
$$|f\rangle = \frac{1}{2}[(\alpha|0\rangle + \beta|1\rangle)|00\rangle + (\alpha|0\rangle - \beta|1\rangle)|01\rangle + (\alpha|1\rangle + \beta|0\rangle)|10\rangle + (\alpha|1\rangle - \beta|0\rangle)|11\rangle]$$

Alice will measure her two qubits in the basis $C = \{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ and complete the transfer of $|\psi\rangle$ to Bob.

Bob will perform some rotation depending on Alice's measurement outcome:

- If Alice measures 00, Bob just receives $|\psi\rangle$ without needing to do anything
- If Alice measures 01, Bob applies a Z gate to the state to get $|\psi\rangle$
- If Alice measures 10, Bob applies a X gate to the state to get $|\psi\rangle$
- If Alice measures 11, Bob applies a Z and X gate to the state to get $|\psi\rangle$





Conclusion and what’s next?

- Classical -> quantum information
  - Notation
  - Qubit representation
  - Control
  - Measurement
- Multi-qubit states and how to generate entangled states
- Components of building quantum circuits

Start working on the beginner challenges and hackathon prompts!